

Signorini's Problem

Max Stuthmann

March 13, 2026

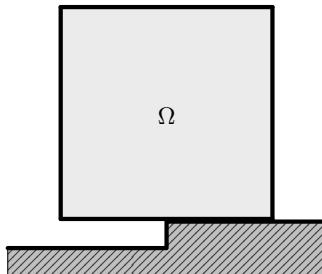
1 Introduction

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- Geometrical and Physical Setup
- Weak Formulations
- Strong Formulations
 - A Hybrid Formulation
 - Formulation with KKT Contact Conditions
 - Classical Strong Formulation under Additional Regularity

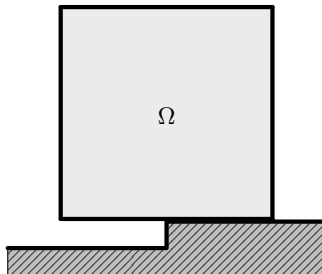
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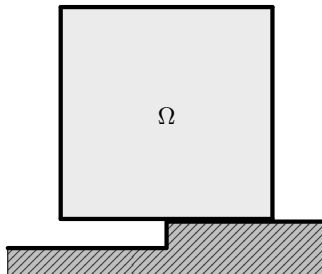
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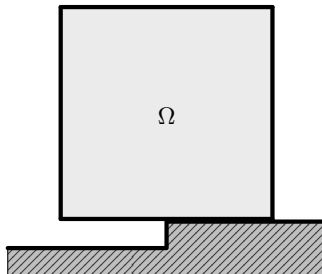
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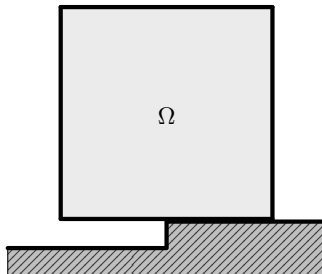
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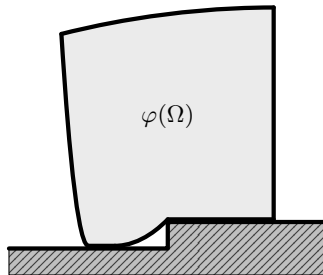
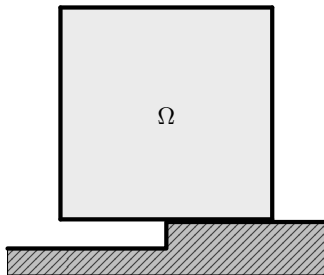
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- ▶ **Not in equilibrium configuration.**



What is to be found?

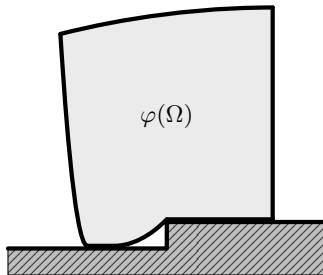
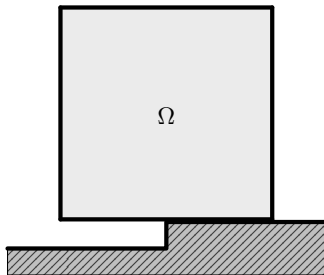
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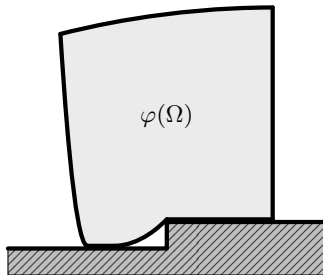
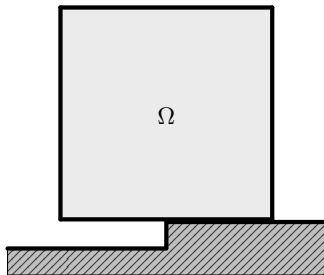


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- ▶ We aim to find the *displacement function* u .



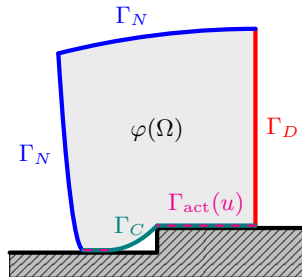
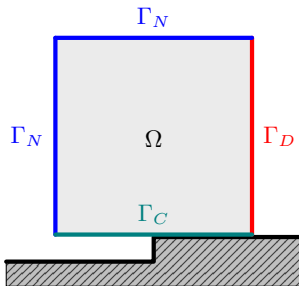
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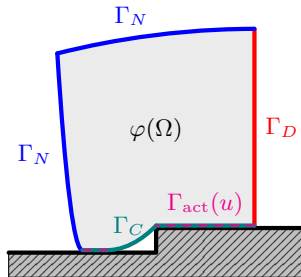
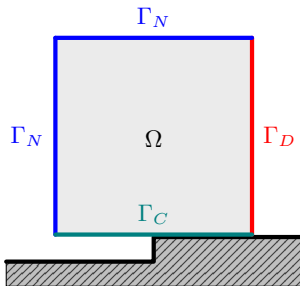
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- Decompose $\Gamma = \partial\Omega$ into three relatively open and pairwise disjoint parts $\Gamma = \overline{\Gamma_D} \dot{\cup} \overline{\Gamma_N} \dot{\cup} \overline{\Gamma_C}$ with $|\Gamma_D|, |\Gamma_C| > 0$.



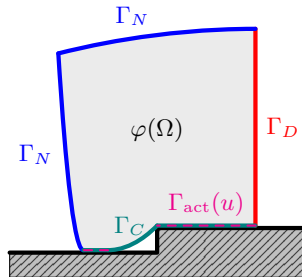
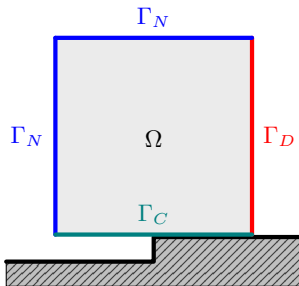
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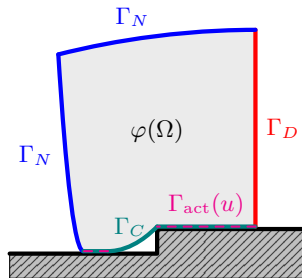
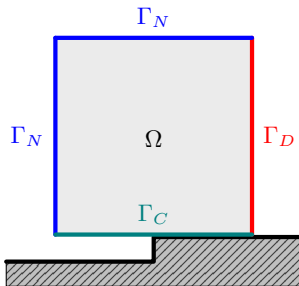
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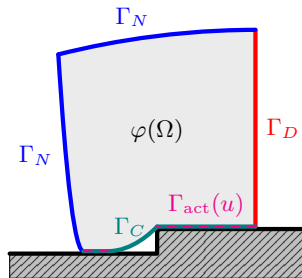
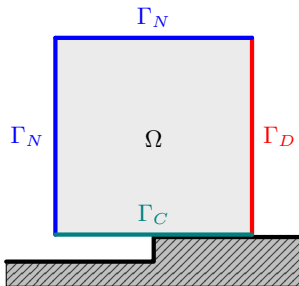
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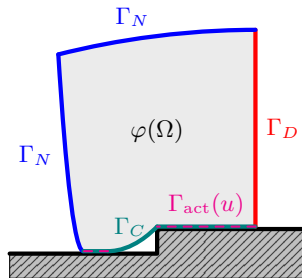
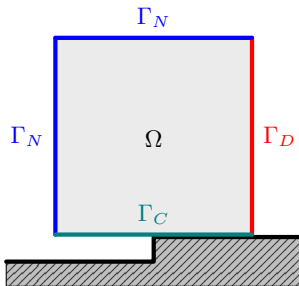
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- ▶ Rigid foundation can be describe by gap-function $g : \Gamma_C \rightarrow \mathbb{R}$. We assume $g = 0$.



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- If $\mathcal{A} : \Omega \rightarrow \mathcal{L}(\mathbb{S}^d; \mathbb{S}^d)$ is a fourth order tensor and $s \in \mathbb{S}^d$, we use the *double contraction*

$$(\mathcal{A}(x) : s)_{ij} := \sum_{k=1}^d \sum_{l=1}^d \mathcal{A}_{ijkl}(x) s_{kl}, \quad s : \mathcal{A}(x) : s := \sum_{i=1}^d \sum_{j=1}^d \sum_{k=1}^d \sum_{l=1}^d \mathcal{A}_{ijkl}(x) s_{ij} s_{kl}.$$

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- ▶ *(Cauchy) stress tensor:* $\sigma : H^1(\Omega; \mathbb{R}^d) \rightarrow L^2(\Omega; \mathbb{S}^d)$, $u \mapsto \mathcal{A} : \varepsilon(u)$.

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- ▶ *Equilibrium* between both: $a(u, v) = \ell(v) \quad \forall v \in H^1(\Omega; \mathbb{R}^d)$.

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- ▶ *Trace Theorem*: Restriction of smooth functions $\overline{\Omega} \rightarrow \mathbb{R}^d$ onto the boundary $\Gamma = \partial\Omega \in \mathcal{C}^{m,1}$ extends uniquely to a continuous surjective linear trace operator

$$\gamma : H^s(\Omega; \mathbb{R}^d) \rightarrow H^{s-1/2}(\Gamma; \mathbb{R}^d), \quad \frac{1}{2} < m < m + 1.$$

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- ▶ K is a *cone*, i.e. if $v \in K$ then $\lambda v \in K$ for all $\lambda \in [0, \infty)$.

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$$a(u, v) = \int_{\Omega} \sigma(u) : \varepsilon(v) \, dx = \int_{\Omega} f \cdot v \, dx + \int_{\Gamma_N} F \cdot (\gamma v) \, ds, \quad \forall v \in V$$

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Problem (Signorini's Problem – Energy Minimization)

Find $u \in K$ *such that:* $u \in \arg \min_{v \in K} \mathcal{J}(v)$

Standard Properties of a and ℓ

Proposition

The symmetric bilinear form $a : V \times V \rightarrow \mathbb{R}$ defined above is

i) *bounded, i.e. for all $v, w \in V$ there exists $C_1 < \infty$ such that*

$$|a(u, v)| \leq C_1 \|u\|_{1,\Omega} \|v\|_{1,\Omega} \quad \text{with} \quad C_1 = C_A,$$

ii) *V -elliptic, i.e. for all $v \in V \setminus \{0\}$ there exist $C_2 > 0$ such that*

$$a(v, v) \geq C_2 \|v\|_{1,\Omega}^2 \quad \text{with} \quad C_2 = \frac{\alpha_A}{C_K^2}.$$

Moreover, the linear functional ℓ defined above is bounded, i.e. for all $v \in V$ there exists $C_3 > 0$ such that

$$|\ell(v)| \leq C_3 \|v\|_{1,\Omega} \quad \text{with} \quad C_3 = \max\{1, C_{\text{tr}}\} (\|f\|_{0,\Omega} + \|F\|_{0,\Gamma_N})$$

Lemma ($\mathcal{J}$ is Coercive and w.l.s.c. on V)

- i) $\mathcal{J}$ is coercive on V , i.e. $\mathcal{J}(v) \rightarrow \infty$ as $\|v\|_{1,\Omega} \rightarrow \infty$.
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Proof:

- By boundedness of ℓ we have $|\ell(v)| \leq \|\ell\|_{V'} \|v\|_{1,\Omega}$. Hence

$$\mathcal{J}(v) = \frac{1}{2} a(v, v) - \ell(v) \geq \frac{\alpha}{2} \|v\|_{1,\Omega}^2 - \|\ell\|_{V'} \|v\|_{1,\Omega} \xrightarrow{\|v\|_{1,\Omega} \rightarrow \infty} \infty.$$

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Lemma (W.l.s.c. of Norms)

Let $(X, \|\cdot\|_X)$ be a normed space. Then $\|\cdot\|_X : X \rightarrow \mathbb{R}_{\geq 0}$ is a weakly lower semicontinuous functional, i.e. if $x_j \rightharpoonup x$ in X , then

$$\|x\|_X \leq \liminf_{j \rightarrow \infty} \|x_j\|_X.$$

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- On $L^2(\Omega; \mathbb{S}^d)$, we define the norm

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- ▶ Since ε is linear and bounded, we have $\varepsilon(v_j) \rightharpoonup \varepsilon(v)$ in $L^2(\Omega; \mathbb{S}^d)$.

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- ▶ $\ell \in V'$ and thus $\ell(v_j) \rightarrow \ell(v)$.
- ▶ Therefore,

$$\mathcal{J}(v) = \frac{1}{2} a(v, v) - \ell(v) \leq \frac{1}{2} \liminf_{j \rightarrow \infty} a(v_j, v_j) - \lim_{j \rightarrow \infty} \ell(v_j) = \liminf_{j \rightarrow \infty} \mathcal{J}(v_j).$$



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The functional $\mathcal{J}$ has at least one minimizer $u \in K$.

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- ▶ Since V is Hilbert space there exists subsequence $\{v_{j_k}\}_{k \in \mathbb{N}}$ and some $u \in V$ such that $v_{j_k} \rightharpoonup u$ in V .

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- ▶ $u \in K$ since K is (weakly) closed.
- ▶ By w.l.s.c. of $\mathcal{J}$, $u \in K$ is minimizer of $\mathcal{J}$ because

$$\mathcal{J}(u) \leq \liminf_{k \rightarrow \infty} \mathcal{J}(v_{j_k}) = M = \inf_{v \in K} \mathcal{J}(v).$$



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If there exists a minimizer of the functional $\mathcal{J}$, it is unique.

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If there exists a minimizer of the functional $\mathcal{J}$, it is unique.

Proof:

- ▶ $\mathcal{J}$ is strictly convex on V and thus on the convex set K . Thus, a minimizer $u \in K$ is unique. $\square$

A-Priori Stability Bound

Proposition (Stability estimate for the minimizer)

The Unique minimizer $u \in K$ of $\mathcal{J}$ satisfies the a-priori bound

$$\|u\|_{1,\Omega} \leq \frac{2 C_K^2 \max\{1, C_{\text{tr}}\}}{\alpha_{\mathcal{A}}} (\|f\|_{0,\Omega} + \|F\|_{0,\Gamma_N}).$$

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- ▶ Finally,

$$\frac{\alpha_{\mathcal{A}}}{2 C_K^2} \|u\|_{1,\Omega}^2 \leq \frac{1}{2} a(u, u) \leq |\ell(u)| \leq \max\{1, C_{\text{tr}}\} (\|f\|_{0,\Omega} + \|F\|_{0,\Gamma_N}) \|u\|_{1,\Omega}.$$



Variational Inequality as an Optimality Condition

Proposition (First-order optimality condition on convex sets)

A function $u \in K$ minimizes $\mathcal{J}$ over K if and only if it satisfies the variational inequality

$$\mathcal{J}'(u)[v - u] = a(u, v - u) - \ell(v - u) \geq 0 \quad \forall v \in K.$$

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- ▶ $\phi(t) := \mathcal{J}(u + t(v - u))$ attains minimum at $t = 0$, i.e. $\phi'(0^+) \geq 0$.
- ▶ By chainrule, $\phi'(0^+) = \mathcal{J}'(u)[v - u] \geq 0$.

Variational Inequality as an Optimality Condition

Proposition (First-order optimality condition on convex sets)

A function $u \in K$ minimizes $\mathcal{J}$ over K if and only if it satisfies the variational inequality

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Find $u \in K$ such that

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- ▶ Normal traces on $H(\operatorname{div}; \Omega)$ are understood as follows.

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- ▶ In the smooth case, $\gamma_n(\tau)$ coincides with the classical traction τn . We thus write $\tau n = \gamma_n(\tau)$.

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- ▶ *Useful identity:*

$$a(u, v) - \ell(v) = \langle \sigma(u)\mathbf{n}, \gamma v \rangle_{\Gamma} - \int_{\Gamma_N} F \cdot (\gamma v) \, ds, \quad \forall v \in V.$$

A Hybrid Formulation

Problem (Signorini's Problem – Hybrid Formulation)

For given $f \in L^2(\Omega; \mathbb{R}^d)$ and $F \in L^2(\Gamma_N; \mathbb{R}^d)$, find $u \in K$ such that $\sigma(u) \in H(\text{div}; \Omega)$ and

$$\begin{cases} -\text{div } \sigma(u) = f & \text{a.e. in } \Omega, \end{cases} \quad (1a)$$

$$\begin{cases} \langle \sigma(u)n, \gamma u \rangle_\Gamma = \int_{\Gamma_N} F \cdot (\gamma u) \, ds, \end{cases} \quad (1b)$$

$$\begin{cases} \langle \sigma(u)n, \gamma v \rangle_\Gamma \geq \int_{\Gamma_N} F \cdot (\gamma v) \, ds \quad \forall v \in K. \end{cases} \quad (1c)$$

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Proposition (Equivalence of Hybrid and Weak Formulation)

Assume $\sigma(u) \in H(\operatorname{div}; \Omega)$. Then $u \in K$ solves the hybrid problem above if and only if it solves the weak problem

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► Let $\varphi \in \mathcal{D}(\Omega; \mathbb{R}^d)$. Then $u \pm t\varphi \in K$ for $|t| \ll 1$, hence the weak inequality implies

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- ▶ This means

$$\langle \sigma(u)n, \gamma v \rangle_{\Gamma} \geq \int_{\Gamma_N} F \cdot \gamma v \, ds + \underbrace{\left[\langle \sigma(u)n, \gamma u \rangle_{\Gamma} - \int_{\Gamma_N} F \cdot \gamma u \, ds \right]}_{=0} \quad \forall v \in V.$$

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$$\gamma_{n,C}(\sigma(u)) = \sigma(u)\mathbf{n} = \sigma_n(u)\mathbf{n} + \sigma_t(u) \in W'_{C,n} \oplus W'_{C,t}.$$

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Problem (Signorini's Problem – Strong Formulation)

For given $f \in L^2(\Omega; \mathbb{R}^d)$, find $u \in V$ such that

$$\left\{ \begin{array}{ll} -\operatorname{div} \sigma(u) = f & \text{a.e. in } \Omega, \\ \sigma_t(u) = 0 & \text{in } W'_{C,t}, \\ u_n \leq 0 & \text{a.e. on } \Gamma_C, \\ \sigma_n(u) \in \Lambda_C, \\ \langle \sigma_n(u), u_n \rangle_{\Gamma_C} = 0. \end{array} \right. \quad \begin{array}{l} (2a) \\ (2b) \\ (2c) \\ (2d) \\ (2e) \end{array}$$

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Classical Strong Formulation under Additional Regularity

Problem (Signorini's Problem – Classical Strong Formulation)

For given $f \in L^2(\Omega; \mathbb{R}^d)$ and $F \in L^2(\Gamma_N; \mathbb{R}^d)$ find $u \in H^s(\Omega; \mathbb{R}^d)$ with $s > 3/2$ such that the small strain elasticity equations

$$\left\{ \begin{array}{ll} -\operatorname{div} \sigma(u) = f & \text{in } \Omega, \\ \sigma(u) = \mathcal{A} : \varepsilon(u) & \text{in } \Omega, \\ u = 0 & \text{on } \Gamma_D, \\ \sigma(u)n = F & \text{on } \Gamma_N, \end{array} \right. \quad \begin{array}{l} (3a) \\ (3b) \\ (3c) \\ (3d) \end{array}$$

and the contact conditions

$$\left\{ \begin{array}{ll} u_n \leq 0 & \text{on } \Gamma_C, \\ \sigma_n(u) \leq 0 & \text{on } \Gamma_C, \\ \sigma_n(u) u_n = 0 & \text{on } \Gamma_C, \\ \sigma_t(u) = 0 & \text{on } \Gamma_C \end{array} \right. \quad \begin{array}{l} (4a) \\ (4b) \\ (4c) \\ (4d) \end{array}$$

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